

Agent-Generated Preference Data for DPO Fine-Tuning — Research

Literature grounding, technique background, and model/architecture selection for a zero-cost DPO fine-tuning project.

Project: Agent-Generated Preference Data for DPO Fine-Tuning of a Small Language Model
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Budget constraint: Every component of this project is designed to run at \$0 cost — free-tier compute (Google Colab T4), free/open-weight models, and free public datasets only.

1. Motivation and Problem Framing

Reinforcement Learning from Human Feedback (RLHF) and its simpler successor, **Direct Preference Optimization (DPO)**, are the dominant techniques used to align language models with human preferences. Both depend on **preference pairs** — a prompt, a "chosen" response, and a "rejected" response. Collecting these pairs from real humans is the single most expensive part of the alignment pipeline — this is why HH-RLHF, the dataset Anthropic released for this purpose, involved large-scale paid human annotation.

This project asks a genuinely open research question: **can a panel of LLM agents, debating and cross-ranking their own answers, generate preference data of high enough quality to meaningfully align a small model via DPO — and how close does the resulting model get to one trained on real human preference data, at zero dollars of cost?** This framing sits inside a broader, actively-studied category called **RLAIF (Reinforcement Learning from AI Feedback)**.

The project is deliberately structured so the agentic/LLM-orchestration layer is a thin data-generation method, while the majority of engineering and intellectual effort sits in the **machine learning core**: model selection, LoRA/DPO training, evaluation design, and ablation analysis.

2. Background: The Techniques Involved

2.1 RLHF vs. DPO

Classic RLHF (established for LLMs by InstructGPT) trains a separate reward model on preference pairs, then uses reinforcement learning (typically PPO) to optimize the policy model against that reward model. This is compute-heavy and notoriously unstable to tune. DPO reframes the same objective as a single closed-form classification-style loss directly on the policy model's log-probabilities of the chosen vs. rejected response, skipping the separate RL loop entirely. This makes DPO dramatically cheaper and more stable to train — which is exactly why it is viable on a single free Colab GPU, where PPO-based RLHF would not be.

2.2 LoRA / QLoRA

Low-Rank Adaptation (LoRA) freezes the base model's weights and trains small rank-decomposed adapter matrices injected into attention/MLP projection layers, cutting trainable parameters and memory by roughly 90% or more. QLoRA adds 4-bit quantization of the frozen base weights, further shrinking memory so that even 7B-class models can be fine-tuned on a single 16 GB T4 GPU. For this project's 1-3B parameter models, LoRA/QLoRA is what makes DPO training feasible on free-tier hardware.

2.3 RLAIF — Why Agent-Generated Preference Data Is a Real Research Angle

The individual pieces are established: Constitutional AI popularized using AI feedback instead of human feedback for harmlessness training, and a direct empirical comparison of AI-generated vs. human preference labels was published as "RLAIF vs. RLHF." What this project adds is a specific, controlled framing — multi-agent debate plus adversarial critic ranking, measured head-to-head against real human-labeled data via ablation on a small model — which is a legitimate small-scale replication-and-extension study, exactly the shape RVCE publication-track projects reward.

3. Base Model Selection — Comparative Research

The single most consequential decision is which small open-weight model to fine-tune. It must (a) run comfortably on a free Colab T4 (16 GB VRAM) with LoRA, (b) have a permissive license, (c) have strong out-of-the-box

instruction-following so DPO is refining behaviour rather than teaching it from scratch, and (d) have first-class support in Hugging Face transformers, trl, and Unsloth.

Model	Params	License	Instruct quality	Notes
Qwen2.5-1.5B-Instruct	1.5B	Apache 2.0	Excellent	Best overall pick. Strong instruction-following for its size, native chat template, broad Unsloth/TRL support, trains comfortably on a T4 in 4-bit.
Llama-3.2-3B-Instruct	3B	Meta Llama Community	Excellent	Stronger raw capability, still fits on a T4 with QLoRA. License is permissive for student/research use but not fully OSI-open.
SmolLM3-3B	3B	Apache 2.0	Very good	Hugging Face's fully transparent model (published data mixture + training recipe) — excellent choice if the report wants to cite an open methodology to compare against.
Gemma-3-1B-IT	1B	Gemma license	Good	Smallest footprint, fastest iteration loop for hyperparameter search; slightly weaker baseline instruction-following than Qwen2.5-1.5B.
Phi-4-mini	3.8B	MIT	Very good	Strong reasoning for its size; heavier than the 1-1.5B options, tighter fit on free T4 with QLoRA required rather than optional.

Recommendation

Primary model: Qwen2.5-1.5B-Instruct, trained with 4-bit QLoRA via Unsloth. It is the best balance of capability, license permissiveness, training speed on a free T4, and ecosystem support. **Secondary/stretch model: Llama-3.2-3B-Instruct or SmolLM3-3B** for a "does scale help" ablation once the 1.5B pipeline is validated end-to-end.

4. Architecture Overview

The system has two layers kept conceptually separate: a **data-generation layer** (agentic, prompt-engineering only, no weights touched) and a **training/evaluation layer** (the actual ML: gradient updates, LoRA adapters, DPO loss, held-out evaluation).

Component	What it does	Tooling
Layer A — Agent Debate	3-4 independent LLM "personas" answer the same prompt.	Free-tier API (Groq/Gemini) or local Ollama model
Layer A — Critic/Ranker	A separate "judge" persona scores pairs of answers against a fixed rubric and outputs chosen/rejected.	Same free-tier API, different prompt
Layer A — Devil's Advocate (optional)	An agent forced to argue the minority position, stress-testing the majority answer.	Same API
Layer B — Preference Dataset	Structured JSONL of {prompt, chosen, rejected} triples.	Local JSONL / HF Datasets
Layer B — Reward Model (optional)	Small classifier over sentence embeddings predicting which response wins.	sentence-transformers + scikit-learn

Component	What it does	Tooling
Layer B — DPO Trainer	LoRA adapters trained with TRL's DPOTrainer against the preference dataset.	transformers + trl + peft + Unsloth
Layer B — Evaluation Harness	Win-rate vs. base model, TruthfulQA/HaluEval subset scoring, reward margin, perplexity.	trl + custom eval scripts

5. Dataset Strategy

- **Seed prompts:** a subset of the HH-RLHF dataset's prompts (prompts only, not its human labels) or the Stanford Alpaca instruction set.
- **Agent-generated labels:** the debate + critic pipeline produces chosen/rejected pairs for those same prompts — the "cheap" preference data.
- **Ground-truth comparison set:** the same prompts' real human-labeled HH-RLHF pairs, kept aside as a reference for the ablation.
- **Held-out benchmarks:** TruthfulQA and HaluEval for evaluation independent of both the agent pipeline and the human-labeled set.

6. Core Research Questions

- 1 Does DPO fine-tuning on agent-generated preference pairs measurably improve the base model's win-rate over its own un-tuned outputs?
- 2 How does a model trained on agent-generated pairs compare, head-to-head, to one trained on an equal-sized sample of real human-labeled HH-RLHF pairs?
- 3 Which agent-pipeline design choices (number of debating agents, presence/absence of the devil's-advocate agent, rubric wording) most affect downstream model quality?
- 4 Does an auxiliary trained reward model's feature importances (via SHAP) reveal what the agent critic is actually rewarding — and does that match a human rubric?

7. Novelty and Publication Angle

The individual pieces (DPO, RLAIF-style AI feedback, LoRA on small models) are all established. The contribution worth writing up is the **controlled ablation**: same base model, same DPO recipe, same evaluation harness, with the only variable being the *source* of preference data (agent-generated vs. human-labeled vs. a naive single-model self-consistency baseline). That three-way comparison, reported with confidence intervals on a public benchmark, is a legitimate short-paper / RVCE publication-track submission.

References & Links

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Meta AI. **Llama-3.2-3B-Instruct — Model Card**. <https://huggingface.co/meta-llama/Llama-3.2-3B-Instruct> — *secondary/stretch base model*

Hugging Face TB Team. **SmolLM3-3B — Model Card**. <https://huggingface.co/HuggingFaceTB/SmolLM3-3B> — *fully transparent alternative base model*

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